

Statistical Models of Horse Racing Outcomes Using R

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Abstract

The published literature on statistical modelling of horse racing outcomes is sparse relative to many other sports, apart from some key seminal works that are now very dated. In addition, there appears to have been little, if any, reference to modelling in horse racing at the MathSport series of conferences since it began as the “IMA International Conference on Mathematics in Sport” in 2007. This lack of published work is not surprising given the potential commercial value of such knowledge and hence the potential unwillingness for authors to share their work.

This paper therefore aims to fill this gap by presenting details of how a potentially profitable model could be developed, using a discrete choice modelling approach within the statistical computing environment R (R Core Team, 2018). Suitable data sources are considered, along with the process of model development and implementation in R, as well assessments of model fit and the potential profitability of the model when used for betting on future races.

1 Introduction

The published literature relating to the development of predictive models in horse racing is relatively sparse, despite it being one of the oldest sports still in common existence today and its long historical relationship with betting. There are many variations of the sport, with two of the most common forms being flat racing (no jumps) and hurdle or steeplechase races over jumps. In addition, races may or may not involve handicapping horses, where the “better” horses are required to carry heavier weights in their saddles to attempt to “even out” the field.

Amongst the seminal published papers in the context of the development of predictive models in horse racing, are Bolton and Chapman (1986), Benter (1994) and Chapman (1994). The first of these, Bolton and Chapman (1986), was the first to outline the use of a multinomial logistic regression model (see Section 3 for details of this model) in the context of 200 handicap races in the USA and utilizing 10 predictor variables. Chapman (1994) extended this work with the multinomial logistic model with a similar much larger database of 2,000 handicap races in Hong Kong, utilizing 20 predictor variables. Both argue that it was possible to derive positive returns from a unit stake betting strategy using a multinomial logistic regression modelling approach. Benter (1994) provides an excellent insight into the potential predictor variables that might be incorporated into an empirical model, and reports on his own successes in developing a profitable system. All three of these papers

appear in the highly recommended publication by Haush et. al. (2008), which is essential reading for anyone interested in developing a predictive model in horse racing, particularly in the context of betting.

A more recent publication is that by Wong (2011), which also includes some reference to the implementation of a multinomial regression model for horse racing in the context of the R programming language (R Core Team, 2018). However, implementing such a model using R presents a number of challenges which are not fully address in that reference. Therefore, this paper aims to provide an outline of how to implement such a model in R, discussing some of the potential problems, as well as possible data sources and the process of developing suitable predictor variables from the data. In addition, suitable model assessments are considered both in relation to model fit using in-sample data and also betting performance when applied to out of sample data.

Section 2 of the paper discusses potential data sources and the data used in the modelling work here, whilst Section 3 then describes the multinomial logistic regression model in the context of horse racing and develops a predictive model using our data. The predictive performance when deployed to new unseen (out of sample) data is then considered in Section 3, with Section 4 providing a summary of the work presented.

2 Data

There is a wealth of available data related to horse racing allows for a rich variety of predictive models to be developed. There are also many data sources that could be considered but many of these are not very helpful. Amongst the more useful data sources are *Proform* (<https://www.proformracing.com/>), *Timeform* (<https://www.timeform.com/horse-racing/>) and *Raceform* (<http://www.raceforminteractive.com/>). These include data on the horses running in each race, the jockeys and the horses' trainers.

The data source we use was compiled from a number of these sources and comes from the Flat Turf Handicaps in the UK, with 16,685 horses taking part in 1,693 races. This data set is obtainable from the author on request and is contained in a .csv file named *horse.data*. This includes for each horse in each race the following variables:

- race.id - unique reference number for each race;
- horse.ref - unique reference number for each horse in each race;
- age - age of the horse (years);
- sireSR - win percentage by offspring of the horse's sire prior to this race;
- trainerSR - win percentage achieved by the horse's trainer prior to this race;
- daysLTO - days since last race (days since Last Time Out);
- position1 - finishing position in the previous race (1, 2, 3 or 4, 0 = anywhere else);
- position2 - finishing position two races ago (1, 2, 3 or 4, 0 = anywhere else);
- position3 - finishing position three races ago (1, 2, 3 or 4, 0 = anywhere else);
- finpos - finishing position in the current race;
- win - indicator of whether each horse won (yes) or not (no);
- sp - starting price obtained from Betfair;
- various other indicators (1=yes, 0=no) of whether the horse was an entire (male horse that has not been castrated), gelding (male horse that has been castrated), and if it was wearing blinkers, a visor, cheekpieces or a tongue tie. Note that a horse that is neither a gelding nor an entire was female.

The variable sp is the starting price obtained from the Betfair betting exchange (<https://www.betfair.com>). This is used here as it represents a kind of consensus of the market just before the event begins, and avoids having to account for the fact that odds change over time. Appendix 1 provides a sample of the data for illustrative purposes. Note that this data set excludes many other variables, such as jockey statistics, but we restrict the number of variables both for brevity, and to maintain the commercial value of the models developed with a fuller version of this data set.

2.1 Data Management and Potential Pitfalls

We split our data (16,685 horses from 1,693 races) into a training sample (70% of races) in order to develop a model and to propose a future possible betting strategy. We retain the remaining 30% of races for out-of-sample assessments of betting performance when the model is applied to new unseen race data. The training set thus consists of 11,710 horses taking part in 1,181 races, with the remaining 4,975 horses from 512 races making up the test set.

Section 3 describes a multinomial logistic regression model developed here, which requires careful consideration of two key variables included in our training data. One of these is `horse.ref`, which provides a means of identifying the horses (or choices) within each race. These only need to be unique within a race as the model provides a means of estimating the probability (more precisely the odds) of winning conditional on the horses it is competing against in that race. The `horse.ref` does not need to be unique to each horse across the data set and in our modelling approach we do not make any connection with horse 1 in race 1 and horse 1 on race 2, for example. The other key variable is `race.id`, which indicates the “choice” groups of horses in each race from which “nature” chooses a winner. The `race.id` must be unique to each race and experience suggests that not checking that this is the case is a common cause of problems, which we consider here.

The model in Section 3 is fitted using the function `mlogit()` from the `mlogit` package of Croissant (2019) within the R statistical computing environment. This requires the data to be in a particular format such that the following function variables need to be specified with care:

- `choice` indicator of which horse won each race
(in our data set this is the variable called `win`);
- `chid.var` variable that defines the choice sets (races) from which the choice of winner is made
(in our data set this is `race.id`);
- `alt.var` variable that defines the choice alternatives (horses) in each set (race)
(in our data set this is `horse.ref`)

The above function variables can be defined in the required format using the `mlogit.data()` function available within the `mlogit` package (where our data set is contained in a data frame in R called `model.data`) as follows:

```
h.dat<- mlogit.data(data=model.data,choice="win",chid.var="race.id",
alt.var="horse.ref",shape="long")
```

The shape is “long” since in our data set we have one row for each horse. We could have included all the horses from one race together on one “wider” row in which case our data shape would have been wide. A common cause of error is with not ensuring the `race.id` (or other variable used to specify the `chid.var` function variable) are unique for each race. Our data set has been checked and “cleaned” to ensure no problems will occur. Resolving issues with this command for other data sets may not always be easy and experience suggests trying running the command on just the first few races in the data set, and increasing the amount of races until the error occurs. This will at least identify which set of race data is the first row to cause problems. Another issue to be wary of is that the `mlogit` package cannot accommodate ties and so any races where two or more horses are tied as winners would need to be removed from the data set.

Within our data set, we also need to consider the issue of the market over-round in the starting prices or odds held in the variable `sp`. These odds do not include the backer’s stake, so for example odds (`sp`) of 1.5 would mean that betting 1 unit on that horse would win 1.5 units and so would see 2.5 units returned to the backer if the horse won. The implied probability (of the horse winning the race) from these odds = $1/(sp+1) = 1/2.5 = 0.4$. Totaling these implied probabilities across all horses in the race should result in a total = 1. However, this total is often greater than 1 and the amount by which this total exceeds 1 is often referred to as the over-round. The histogram

in Figure 1 below provides a summary of the over-rounds evident in our training data set and shows that this is mostly around the 1.1 or 1.2 mark, but can be as much as 1.5.

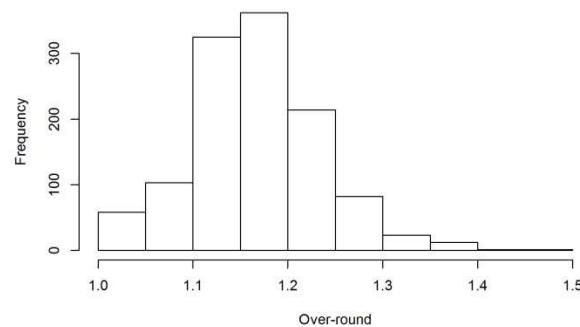


Figure 1: Histogram summary of the over-rounds within the training data set

In order to make more informed assessments of the model versus the market odds, and also to make better judgements with respect to betting strategies, we derive to additional variables to add to our data set. The first of these we name `market.prob`, which contains the market implied probabilities of each horse winning any particular race. The other we call `market.prob.adjusted`, which contains the market implied probabilities of each horse winning any particular race but adjusted by dividing by the over-round for that race so that those probabilities do sum to 1.

`SireSR` and `trainerSR` are, respectively, the strike rate (percentage of races won) by the offspring of the horse's Sire (father) prior to this race and the win percentage achieved by the horse's trainer prior to this race. The boxplots in Figure 2 summarise the data for both `sireSR` and `trainerSR` in the training data set. In both cases, the vast majority have `sireSR` and `trainerSR` values of less than 20% and only a handful of horses have values above this. A value above 20% would also be considered to be very unusual. Hence for both variables, we censor the values so that any above 20% instead take the value of 20%.

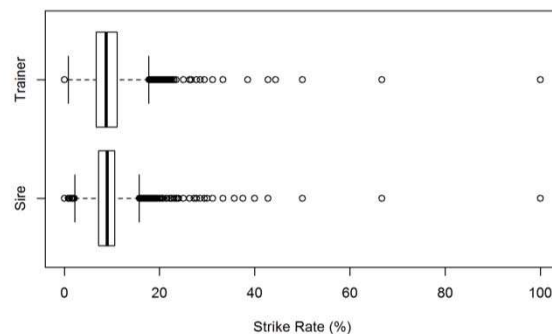


Figure 2: Boxplot summaries of `sireSR` and `trainerSR` within the training data set

Examining `daysLTO`, which is the number of days since the horse's last race (days since Last Time Out), lead us to decide to also censor this variable so that values of `daysLTO` above 60 days are capped at 60 days. For brevity, the data summaries used to support this decision are not included here.

2.2 Exploratory Analysis

This section explores the training data set, and aims to provide an illustration of some of the types of initial assessments that need to be undertaken before any attempt is made to develop a model. One of the key ingredients in a successful model, is the development of new more meaningful variables from those that are collected within the original data set. A knowledge of horse racing and how various factors could affect a horse's performance is therefore crucial in this work.

We first consider age. Gramm and Marksteiner (2011) offers evidence that a horse's peak age for racing is around 4.5 years. These findings are supported by the results in Figure 3, which summarises the proportion of winning horses in each age category (in years) in our training data set.

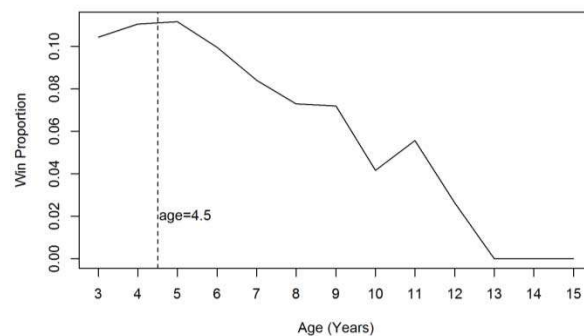


Figure 3: Win proportion versus Age

It would therefore seem sensible to define a revised version of the variable age that might serve as a better predictor of the odds of winning. Here, the absolute difference in age from 4.5 years is utilized instead and referred to as *age.diff*. Note that this assumes the rate of improvement from age 3 to 4.5 is the same as the rate of decline after age 4.5, which would seem to be supported in Figure 3. However, Gramm and Marksteiner (2011) found that the improvement from age 3 to 4.5 is greater than the rate of decline after age 4.5.

The conditional density plots in Figure 4 summarise the win proportion against *sireSR* and *trainerSR*. Conditional densities are used to smooth out random fluctuations in win proportion observed with increasing strike rate. Both plots show an increase in win proportion with higher values of *sireSR* and *trainerSR* and hence evidence their values as potential predictors in the model.

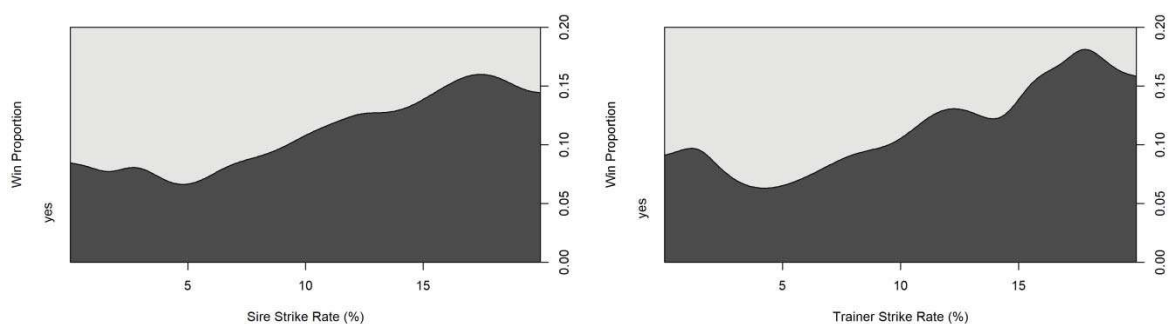


Figure 4: Win proportion versus Sire Strike Rate and Trainer Strike Rate

The conditional density plot in Figure 5, similarly summarises the win proportion against daysLTO, and illustrates how win proportion decreases with higher values of daysLTO.

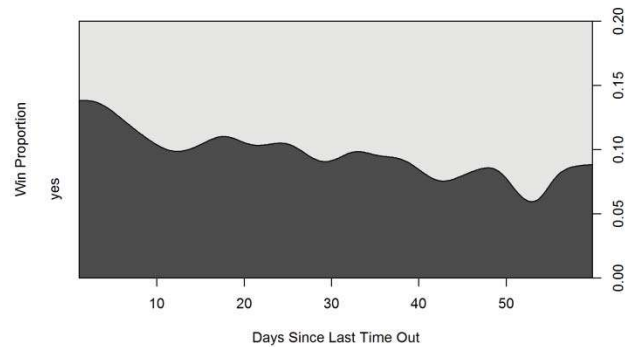


Figure 5: Win proportion versus Days Since Last Time Out

Figure 6 below summarises how win proportion varies with a horse's finishing position in its previous race, two races previous and three races previous. This suggests that higher finishing positions in the previous race and two races previous (variables position1 and position 2 in our data set) are associated with a greater win proportion in the subsequent race. The result three races previous does not seem to have much impact on the current race outcome.

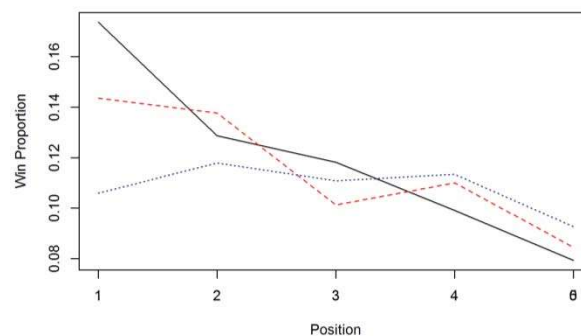


Figure 6: Win proportion versus Position in the horse's previous race (—), two races previous (---) and three races previous (.....)

Finally, Table 1 summarises the remaining categorical variables in relation to win proportion. The first row indicates the win percentages for horses that were entire, geldings, or wore blinkers, visor, cheekpieces or a tongue tie. The second row indicates the win proportions for horses that were not entire, or not geldings or did not wear the item listed. This suggests that greater win proportions in the sample are evident with horses that were entire and geldings (i.e. male horses) compared to female horses, and also with horses that were wearing blinkers or not wearing cheek pieces or a tongue tie. The impact of a visor across the sample seems minimal in terms of win proportion.

	Entire	Gelding	Blinkers	Visor	Check Pieces	Tongue Tie
Yes	0.115	0.106	0.111	0.103	0.069	0.084
No	0.099	0.091	0.100	0.101	0.103	0.102

Table 1: Win proportion versus the remaining categorical predictor variables

3 Model and Results

3.1 Model Definition and Fitting Using R

The model we utilise is the multinomial logistic regression model. There are actually a number of different forms of the multinomial logistic regression model and they are often collectively referred to as discrete choice models. In our context we treat each horse in a race as the set of choices from which nature will choose a winner when the race is run. The multinomial model is usually framed in the context of people making choices and so we often see references to individual-specific variables and alternative-specific variables. In our case the race is making the choice of horse and so individual-specific variables are the race-specific variables and the alternative-specific variables are the horse-specific variables. This is often the source of confusion that prevents many implementing the multinomial logistic regression model in the context of horse racing.

For the multinomial logistic regression model we define linear predictor terms V_i for horse i as:

$$V_i = \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip}$$

where $x_1, x_2, \dots, x_p$ are the values of the horse-specific variables (age, sireSR, trainerSR etc.) for horse i and $\beta_1, \beta_2, \dots, \beta_p$ are the associated model parameters to be estimated in the model. We could also extend this to include other variables that are race-specific if required, such as the going (the condition of the course) or the value (standard) of the race. However, our data set does not include such variables and so we ignore this aspect of the model here. The model can then be stated in terms of the probability P_i that horse i will win the race as follows:

$$P_i = \frac{\exp(V_i)}{\sum_{i=1}^n \exp(V_i)},$$

where there are n horses in the race. Note that we have no intercepts (β_0 terms) in the model, since all we are interested in are the relative chances of each horse winning relative to all the other horses in the same race. The above model also has the useful property that for any given race, $\sum_{i=1}^n P_i = 1$.

Fitting the model with all of our predictor variables in the model is then undertaken using the `mlogit()` function as follows:

```
mlogit(win~age.diff+sireSR+trainerSR+daysLTO+position1+position2+position3+entire
+gelding+blinkers+visor+cheekpieces+tonguetie|0|0,data=h.dat)
```

The two vertical lines after `tonguetie` are used to separate the predictor variables into three parts. The first part (before the first of the two vertical lines) contains the predictor variables that are horse-specific, which in our case is the list of all of our predictor variables. The second part (between the two vertical lines) which currently has just a zero listed, would contain the predictor variables that are race-specific, but we have none of these and so we state this using 0. The third part (after the last of the two vertical lines) which also currently has just a zero listed, can be used to list the predictor variables that are again horse-specific (like the first part), but in this case the way in which these variables affect the win probability would be allowed to vary from horse to horse. Again, since we have none of these we simply state this by using 0.

3.2 Model Results

The results of fitting an initial model with all predictor variables in the model are shown in Table 2. The reference categories for the three "Position" variables was "0" (i.e. unplaced in the previous races).

Parameter	Estimate	Std. Error	p
age.diff	-0.153	0.0314	<0.001
sireSR	0.048	0.0093	<0.001
trainerSR	0.051	0.0093	<0.001
daysLTO	-0.004	0.0018	0.020
Position1	1	0.602	0.0919
	2	0.324	0.1006
	3	0.312	0.1027
	4	0.159	0.1082
Position2	1	0.368	0.0974
	2	0.363	0.0982
	3	0.066	0.1074
	4	0.213	0.1050
Position3	1	-0.046	0.1061
	2	0.109	0.1000
	3	0.117	0.1036
	4	0.130	0.1013
entire	0.499	0.1297	<0.001
gelding	0.557	0.0948	<0.001
blinkers	0.016	0.1125	0.89
visor	0.027	0.1443	0.85
cheekpieces	-0.504	0.1470	0.001
tonguetie	-0.297	0.1632	0.069

Table 2: Parameter estimates for an initial model including all predictor variables

This suggests the following are all highly statistically significant ($p < 0.001$) predictors of the odds of winning: age.diff, sireSR, trainerSR, daysLTO, position1, position2, entire, gelding and cheekpieces. In addition, tonguetie is borderline not significant ($p = 0.069$) but we chose to exclude this. There was no evidence that the following should be retained in the model: position3 ($p = 0.43$), blinkers ($p = 0.89$) and visor ($p = 0.85$).

A process of developing a model by removing the most non-significant predictor variable above (i.e. starting with the removal of blinkers) was undertaken until only significant predictors remained. The resulting model is summarised in Table 3. Note that the exclusion of visor was a borderline decision and the decision to exclude this was argued on the basis that the more parsimonious model is preferable. The results in Table 3 indicate that the odds of winning are increased for male horses (entires and geldings) compared to female horses. Increased odds are also associated with horses closer to the optimal age (4.5 years) and who have higher values for sireSR and trainerSR, lower values for daysLTO and a higher placing (1st being highest) in the previous two races. In addition, the wearing of cheek pieces is also seen to decrease the odds of winning.

Parameter		Estimate	Std. Error	P
age.diff		-0.152	0.0313	<0.001
sireSR		0.048	0.0093	<0.001
trainerSR		0.051	0.0093	<0.001
daysLTO		-0.004	0.0018	0.021
Position1	1	0.607	0.0918	<0.001
	2	0.329	0.1003	
	3	0.313	0.1026	
	4	0.157	0.1081	
Position2	1	0.377	0.0970	<0.001
	2	0.373	0.0975	
	3	0.075	0.1070	
	4	0.220	0.1048	
entire		0.489	0.1294	<0.001
gelding		0.548	0.0946	<0.001
cheekpieces		-0.503	0.1454	0.001

Table 3: Parameter estimates for a "final" model

3.3 Model Fit Assessments

One key assessment of model fit is that the model should be well calibrated. That is, for example, on average, horses with a model win probability of 0.2 should win 20% of races they take part in. The horses in the training set were therefore grouped into 9 sets (bins) based on whether their model fitted win probabilities fell into the following intervals: $[0,0.05]$, $(0.05,0.1]$, ..., $(0.35,0.40]$, $(0.4,1.0]$. Figure 7 plots the mean value for the model fitted win probabilities for each of these intervals and the associated win proportions that were actually observed (plotted using the symbol "o"). The plot also shows the same results for the win probabilities implied from the market odds adjusted for over-round (+). The closer the points are to the reference line shown the more accurate the fitted probabilities are on average, and hence are better calibrated. This suggests the model is well calibrated although less so for the higher win probabilities, but the sample size for these higher win probabilities is much smaller being subject to more random fluctuation. Figure 7 would also seem to suggest that the model performs very well compared to the market implied probabilities.

Another potentially useful measure of model performance are those related to scoring rules used for prediction, although in this case we are considering in-sample predictions. Two such measures we consider here are $P1$ and $P2$ as follows:

$$P1 = \exp \left\{ \frac{1}{N} \sum_k \log(P_{jk}) \right\}$$

where P_{jk} represents the model fitted win probability (or win probability implied by the odds adjusted for over-round) for horse j that was observed to win race k , for races $k=1, \dots, N$, and

$$P2 = \exp \left\{ \frac{1}{N} \sum_k \left[\log(1 - P_{jk})^2 + \sum_{i \neq j}^{n_k} \log(P_{ik})^2 \right] \right\}$$

where P_{jk} represents the model fitted win probability (or win probability implied by the odds adjusted for over-round) for horse j that won race k . Here, P_{ik} represents the model fitted win probabilities (or win

probability implied by the odds adjusted for over-round) for all other horses $i \neq j$, that did not win race k , for horses $i=1, \dots, n_k$ in race k , for $k=1, \dots, N$.

In effect, $P1$ is equivalent to the geometric mean of the model fitted win probabilities for the horses that won the races in the training set. Hence, higher values of $P1$ are preferable. In contrast, $P2$ represents a form of quadratic loss or brier scoring rule, and as such measures the squared error in the model, and so smaller values are preferable.

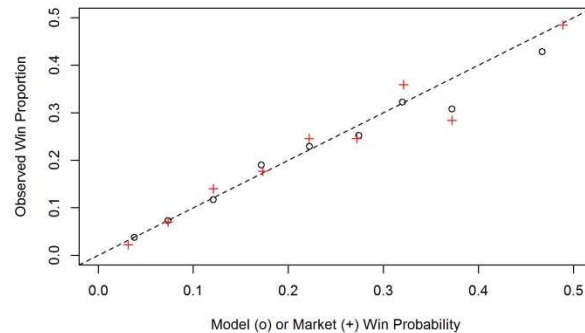


Figure 7: Calibration plot for the model (o) and market implied win probabilities (+)

Calculating $P1$ and $P2$ for the training set, for both the model fitted win probabilities and the win probabilities implied by the market odds adjusted for over-round, gave $P1_{\text{model}} = 0.120$ and $P1_{\text{market}} = 0.136$, and $P2_{\text{model}} = 0.860$ and $P2_{\text{market}} = 0.831$. These suggest the market outperforms the model overall. However, this is not the full story as shown in Figure 8. This plots the values of $P1_{\text{model}}$ compared to $P1_{\text{market}}$ and $P2_{\text{model}}$ compared to $P2_{\text{market}}$, but based only on horses with win probabilities above the values shown on the x-axis, which ranged from 0 to 0.5. For example, based only on horses where the model fitted win probabilities exceed 0.13, we see that $P1_{\text{model}} = P1_{\text{market}}$ and $P2_{\text{model}} = P2_{\text{market}}$. Therefore, if for example we focus only on horses where the model fitted win probabilities exceed, say, 0.15, we see that $P1_{\text{model}} > P1_{\text{market}}$ and $P2_{\text{model}} < P2_{\text{market}}$ and so in that case the model outperforms the market. One possible reason for this could be due to the well-established favourite-long-shot bias. In this case, the odds of long-shots are lower than those that might reflect "fair" odds and so the implied probabilities are higher than would be expected and hence biased. In contrast, the odds for favorites (or horses with higher win probabilities) are actually relatively higher than expected, resulting in lower implied win probabilities. These results suggest there is scope using this model, betting only in horses with a model win probability above some threshold such as 0.15.

A naïve betting strategy to deploy to the so far unseen test data, might therefore to simply choose to bet on horses which has a model win probability greater than 0.15 (established above). In addition, if we consider the model win probability implying "fair" odds (often referred to as an odds-line), then we might wish to also restricts bets to where the model win probability is greater than that implied from the market odds (sp). However, experience with developing betting strategies of this nature, suggests that this may not represent an optimal strategy. A more informed strategy would be to bet where the ratio of model win probability / market implied win probability is greater than some threshold (greater than 1). Further analyses based on the training data set again only (not shown here for brevity) suggested that betting when the ratio of the win probabilities from the model/market is above 1.3 appeared optimal.

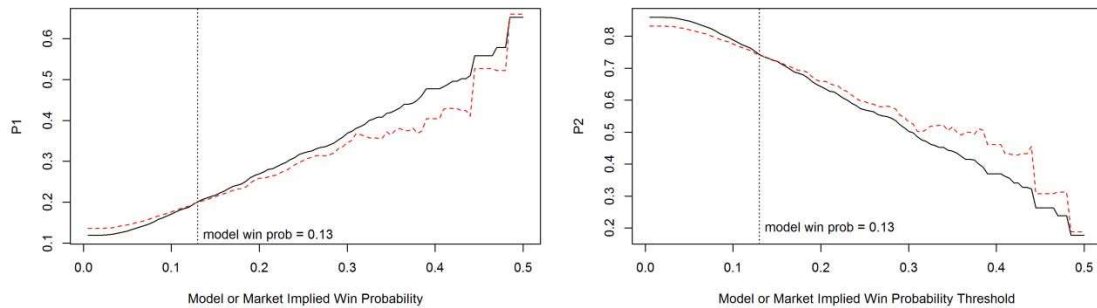


Figure 8: Plots of $P1$ and $P2$ based on win probabilities above the values shown on x-axis for Model (—) and the Market implied win probabilities (---)

3.4 Future Predictive Performance

The model summarised in Table 3 was deployed to the so far unseen test data set (4,975 horses from 512 races) to derive estimated win probabilities for each horse in those races. Unit bets were placed "virtually" on those horses where the model win probability was greater than 0.15 and the ratio of the win probabilities from the model/market was above the threshold of 1.3. This returned a profit of 54 units betting on 264 horses, which represents a positive return on investment (ROI) of around 20% (return of 54 units from 264 units staked). Clearly this might be due to chance and so in Figure 9 we summarise the potential variation in this return using bootstrap estimates of the standard error. This shows the potential ROI when deploying the model using a betting strategy adopting a range of thresholds for the ratio of the win probabilities from the model/market (from 0.9 to 2.0), and includes the 90% confidence interval for this estimate. This provides evidence to suggest that whilst positive returns are not guaranteed, they would appear to have been quite likely using our model and betting strategy.

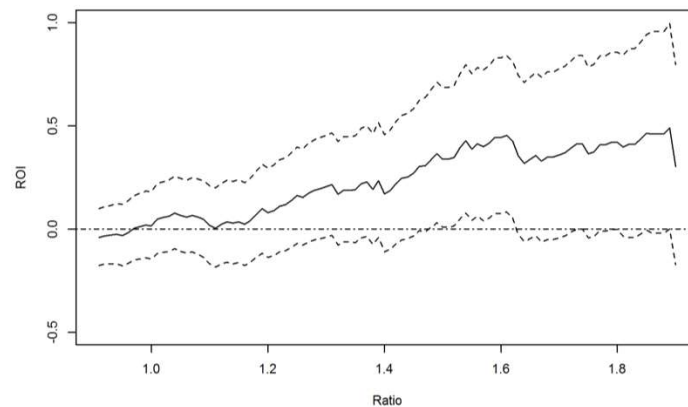


Figure 9: ROI when betting above a range of thresholds of model win probability/market implied win probability (—) and 90% Confidence Intervals (---)

4 Summary

This paper has shown how a potentially profitable model could be developed, using a discrete choice modelling approach within the statistical computing environment R. A large data set with 16,685 horses taking part in 1,693 races from the Flat Turf Handicaps in the UK was utilised. A model was developed from a training set representing 70% of the races (11,710 horses taking part in 1,181 races). with the remaining 4,975 horses from 512 races making up a test set. A multinomial logistic regression model was developed which was shown to be well-calibrated and competitive in relation to the market prices contained within the Betfair starting prices. A betting strategy, again developed using only the training set, was deployed to with the model using the test data. The results suggest that positive returns are possible using such an approach. The data set and R code used in the analysis are both available from the author on request.

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